

## **Step 1: Identify Normalization Issues**

### **➤ 1NF (First Normal Form) – Atomicity**

Problem:

- author\_names contains multiple values (Alice Smith, Bob Jones)
- categories contains multiple values (Fiction, Drama)
- Each field should hold only one value per record.

To fix: Split into separate tables so that each author and category is stored as a single value per row.

### **➤ 2NF (Second Normal Form) – Partial Dependency**

Problem:

- The table's primary key could be (member\_id, book\_id) (since a member can borrow multiple books).
- Non-key attributes like member\_name, book\_title, publisher\_name, etc. depend only on part of the key, not the whole.

To fix:

- Separate data into Members, Books, and Borrowing tables so each attribute depends on its entity.

### **➤ 3NF (Third Normal Form) – Transitive Dependency**

Problem:

- publisher\_country depends on publisher\_name, not directly on the book or borrowing.
- fine\_amount depends on borrowing (a derived or dependent field)
- To fix:
- Move publisher info to a Publishers table.
- Move fine info to a Fines table.

- **Step 2: Normalize Step by Step**

## 1NF → Split multi-valued fields

We create separate rows for authors and categories.

New tables:

- Authors (for individual authors)
- Categories (for individual categories)

## 2NF → Separate entities

We separate members, books, and borrowings.

Tables after 2NF:

- Members(member\_id, member\_name)
- Books(book\_id, book\_title, publisher\_name)
- Borrowing(member\_id, book\_id, borrow\_date, return\_date, fine\_amount)

## 3NF → Remove transitive dependencies

Move publishers and fines to their own tables.

Tables after 3NF:

- Members(member\_id, member\_name)
- Publishers(publisher\_id, publisher\_name, publisher\_country)
- Books(book\_id, book\_title, publisher\_id)
- Authors(author\_id, author\_name)
- BookAuthors(book\_id, author\_id)
- Categories(category\_id, category\_name)
- BookCategories(book\_id, category\_id)
- Borrowing(borrow\_id, member\_id, book\_id, borrow\_date, return\_date)
- Fines(fine\_id, borrow\_id, fine\_amount)

### Step 3: Explain the Improvements

| Normal Form | Problem Solved                       | Benefit                                                       |
|-------------|--------------------------------------|---------------------------------------------------------------|
| 1NF         | No repeating or multi-valued columns | Easier to query, update, and maintain                         |
| 2NF         | Removes partial dependencies         | Prevents duplicate member/book info                           |
| 3NF         | Removes transitive dependencies      | Ensures each fact stored only once; prevents update anomalies |

### Step 4: Final Normalized Tables (3NF)

#### ❖ Members

| member_id | member_name |
|-----------|-------------|
| 101       | John Doe    |

#### ❖ Publishers

| publisher_id | publisher_name | publisher_country |
|--------------|----------------|-------------------|
| 1            | ABC Press      | USA               |

#### ❖ Books

| book_id | book_title      | publisher_id |
|---------|-----------------|--------------|
| B001    | The Great Novel | 1            |

#### ❖ Authors

| author_id | author_name |
|-----------|-------------|
| 1         | Alice Smith |
| 2         | Bob Jones   |

#### ❖ BookAuthors

| book_id | author_id |
|---------|-----------|
|---------|-----------|

| <b>d</b> |   |
|----------|---|
| B001     | 1 |
| B001     | 2 |

### ❖ Categories

| <b>category_id</b> | <b>category_name</b> |
|--------------------|----------------------|
| 1                  | Fiction              |
| 2                  | Drama                |

### ❖ BookCategories

| <b>book_id</b> | <b>category_id</b> |
|----------------|--------------------|
| B001           | 1                  |
| B001           | 2                  |

### ❖ Borrowing

| <b>borrow_id</b> | <b>member_id</b> | <b>book_id</b> | <b>borrow_date</b> | <b>return_date</b> |
|------------------|------------------|----------------|--------------------|--------------------|
| 1                | 101              | B001           | 2025-10-01         | 2025-10-20         |

### ❖ Fines

| <b>fine_id</b> | <b>borrow_id</b> | <b>fine_amount</b> |
|----------------|------------------|--------------------|
| 1              | 1                | 5.00               |